- `Person(pid, firstName, lastName, email, affiliation, startDate, endDate)`

$$\text{pid} \longrightarrow \text{firstName}, \text{lastName}, \text{email}, \text{affiliation}, \text{startDate}, \text{endDate}$$

- `Student(pid, program)`

$$\text{pid} \longrightarrow \text{program}$$

- `Employee(pid, phone, office, supervisorPid)`

$$\text{pid} \longrightarrow \text{phone}, \text{office}, \text{supervisorPid}$$

- `Academic(pid)`

$$\text{pid} \longrightarrow \text{pid}$$

- `nonAcademic(pid)`

$$\text{pid} \longrightarrow \text{pid}$$

- `Faculty(pid, position)`

$$\text{pid} \longrightarrow \text{position}$$

- `Administrative(pid, position)`

$$\text{pid} \longrightarrow \text{position}$$

- `Technical(pid, position)`

$$\text{pid} \longrightarrow \text{position}$$

- `Advises(academicPid, studentPid)`

$$\text{academicPid}, \text{studentPid} \longrightarrow \text{academicPid}, \text{studentPid}$$

- `Laboratory(labId, name, building, roomNumber, discipline, supervisorPid)`

$$\text{labId} \longrightarrow \text{name}, \text{building}, \text{roomNumber}, \text{discipline}, \text{supervisorPid}$$

- `Attached(pid, labId)`

$$\text{pid}, \text{labId} \longrightarrow \text{pid}, \text{labId}$$

- `ResearchProject(code, title, startDate, endDate, status)`

$$\text{code} \longrightarrow \text{title}, \text{startDate}, \text{endDate}, \text{status}$$

- `Participates(pid, projectCode, role)`

$$\text{pid}, \text{projectCode} \longrightarrow \text{role}$$

- `Budget(budgetLine, amountGranted, amountDisbursed, startDate, endDate, managerPid)`

$$\text{budgetLine} \longrightarrow \text{amountGranted}, \text{amountDisbursed}, \text{startDate}, \text{endDate}, \text{managerPid}$$

- `FundsPrj(budgetLine, projectCode)`

$$\text{budgetLine}, \text{projectCode} \longrightarrow \text{budgetLine}, \text{projectCode}$$

- `FundsLab(budgetLine, labId)`

$$\text{budgetLine}, \text{labId} \longrightarrow \text{budgetLine}, \text{labId}$$

- `EquipmentModel(modelId, commercialName, manufacturer, category, requiredEnvironment, trainingManda`

$$\text{modelId} \longrightarrow \text{commercialName}, \text{manufacturer}, \text{category}, \text{requiredEnvironment}, \text{trainingMandatory}$$

- `EquipmentUnit(serialNo, acquisitionDate, purchaseCost, status, portable, modelId, labId)`

$$\text{serialNo} \longrightarrow \text{acquisitionDate}, \text{purchaseCost}, \text{status}, \text{portable}, \text{modelId}, \text{labId}$$

- `Certification(code, title, issuingAuthority, validityPeriod, safetyLevel)`

$$\text{code} \longrightarrow \text{title}, \text{issuingAuthority}, \text{validityPeriod}, \text{safetyLevel}$$

- `Requires(modelId, certCode)`

$$\text{modelId}, \text{certCode} \longrightarrow \text{modelId}, \text{certCode}$$

- `Holds(pid, certCode, issueDate, expirationDate, grade)`

$$\text{pid}, \text{certCode} \longrightarrow \text{issueDate}, \text{expirationDate}, \text{grade}$$

- `Reservation(resId, submissionTS, plannedStart, plannedEnd, purpose, status, madeByPid, projectCode approverPid)`

$$\text{resId} \longrightarrow \text{submissionTS}, \text{plannedStart}, \text{plannedEnd}, \text{purpose}, \text{status}, \text{madeByPid}, \text{projectCode}, \text{approverPid}$$

- `Reserves(resId, serialNo)`

$$\text{resId}, \text{serialNo} \longrightarrow \text{resId}, \text{serialNo}$$

- `Maintenance(serialNo, startTS, endTS, type, description, cost, outcome, technicalPid)`

$$\text{serialNo}, \text{startTS} \longrightarrow \text{endTS}, \text{type}, \text{description}, \text{cost}, \text{outcome}, \text{technicalPid}$$

- `CalibrationRecord(serialNo, calibDate, calibrationType, result, nextDueDate, remarks)`

$$\text{serialNo}, \text{calibDate} \longrightarrow \text{calibrationType}, \text{result}, \text{nextDueDate}, \text{remarks}$$

- `Consumable(consId, name, unitOfMeasure, hazardLevel, reorderThreshold)`

$$\text{consId} \longrightarrow \text{name}, \text{unitOfMeasure}, \text{hazardLevel}, \text{reorderThreshold}$$

- `Supplier(suppId, name, contactEmail, phone)`

$$\text{suppId} \longrightarrow \text{name}, \text{contactEmail}, \text{phone}$$

- `Supplies(suppId, consId, labId, unitPrice)`

$$\text{suppId}, \text{consId}, \text{labId} \longrightarrow \text{unitPrice}$$

- `Stocks(labId, consId, quantityOnHand, lastRestockDate, storageCondition, technicalPid, monitoringS`

$$\text{labId}, \text{consId} \longrightarrow \text{quantityOnHand}, \text{lastRestockDate}, \text{storageCondition}, \text{technicalPid}, \text{monitoringSince}$$

- `Consumes(resId, labId, consId, quantityUsed)`

$$\text{resId}, \text{labId}, \text{consId} \longrightarrow \text{quantityUsed}$$